

BE A WARDEN

Project Team

Bilal Ur Rehman	21I-0472
Mahad Saleem	21I-0475
Talha Zahoor	21I-0867

Session 2021-2025

Supervised by

Mr. Shams Farooq

Co-Supervised by

Dr. Muhammad Asim



Department of Computer Science

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

June, 2025

Contents

1	Introduction	1
1.1	Problem Statement	1
1.2	Scope	2
1.3	Modules	2
1.3.1	Module 1	2
1.3.2	Module 2	3
1.4	User Classes and Characteristics	3
2	Project Requirements	5
2.1	Use-case Table	6
2.2	Functional Requirements	7
2.2.1	Module 1: App Development & License Plate Detection	7
2.2.2	Module 2: Illegal Number Plate Detection & HTV First Lane De- tection	8
2.3	Non-Functional Requirements	8
2.3.1	Reliability	8
2.3.2	Usability	9
2.3.3	Performance	9
2.3.4	Security	9
2.4	Domain Model	10
3	System Overview	11
3.1	Architectural Design	11
3.2	Design Models	12
3.2.1	Activity Diagram	12
3.2.2	System Sequence Diagram	13
3.2.3	Data Flow Diagram	15
3.3	Data Design	15
3.3.1	Databases	15
	References	17

List of Figures

2.1	Domain Model Diagram	10
3.1	Architectural Design Diagram	11
3.2	Activity Diagram	12
3.3	System Sequence Diagram	13
3.4	Data Flow Diagram	15

List of Tables

1.1	User Classes and Characteristics	3
2.1	Use Case	6

Chapter 1

Introduction

"Be A Warden" is an app that uses AI to detect violations and enforce traffic rules . We allow specific users to get registered to our system. They then record videos through our app and upload to our system which then uses AI models to detect multiple violations and notify the authorities which will then use e-challan.

Our project mainly focuses on introducing dynamic cameras and reduce human effort and capture violations missed by the traffic wardens and static cameras.

1.1 Problem Statement

Increasing Traffic Violations: Traffic violations, such as speeding, illegal lane changing and broken brake lights are becoming more frequent, leading to unsafe roads and increasing accident rates. People tend to violate traffic rules where there are no cameras or wardens.

Limitations of Static Cameras: Cameras fixed on roads to detect traffic violations are limited to a place and can detect traffic violations that are happening at that location. There are a lot of places where there are no cameras and people violate traffic rules there and thus this is a limitation of static camera.

Insufficient Human Resources: Traffic wardens are limited and can cover limited areas to check traffic violations and supervise traffic and a warden cant detect every traffic violation by every violator. It is difficult for a warden to supervise hundreds or cars at a time and check every person.

Need for Community Involvement: There is a lack of engagement from the general public in reporting violations. General public doesn't get involved in reporting traffic violations as they aren't getting any benefit from it.

Increasing Accidents: Traffic violations cause a lot of accidents. Every year there are roughly 10 thousand accidents of which 50 percent are fatal and this percentage is growing every year. Strict traffic supervision can control these accidents.

1.2 Scope

The scope of the project basically focuses on providing an AI-powered traffic violation system that enables registered users to act as virtual wardens. Our users will be able to record certain violations through our app. They will become registered users or virtual wardens after passing a test. Our App will provide an interface to record violations through their mobile phones fixed in the car. The videos will be sent to centralized server for violation detection through AI. Then using licence plate recognition system the violator will be tracked and the authorities will be notified and then a challan will be sent to the violator by the authorities. Basically this system is providing help by utilizing dynamic cameras used by virtual wardens. The system will cover a wide range of violations described in the modules. The data privacy will be taken care of. Future expansion possibilities include advanced AI algorithms to detect complex traffic behaviors and also integration of our system with smart city cameras and system.

1.3 Modules

Write down the modules of the proposed project. Each module should highlight features, using bulleted/numbered notation. When developing both a mobile app and a web app, group the modules according to the system types, such as, Client Web App, Client Mobile App, Admin Web App etc.

1.3.1 Module 1

App Development

1. Camera Recording (Using our app to record violations occurring on the specified roads.)
2. Voice Command Recognition (Use voice command in app to auto record our videos.)
3. UI/UX

Licence Plate Detection

1. OCR Number Plate Detection (Use OCR to get number plates of the cars)

1.3.2 Module 2

Illegal Number Plate Detection

1. Custom Dataset Collection (Collect dataset for Pakistan Number Plates Of Each region)
2. Annotation (Label Dataset)
3. Model Training (Train model to detect illegal number plates and also number plates of each region)

HTV First Lane Detection

1. First Lane Detection (Detect HTV in First Lane)
2. Vehicle Classification (Classify each type of vehicle like HTV or LTV)
3. Model Training (Train model to detect HTV and LTV)

1.4 User Classes and Characteristics

User Class	Description
Virtual Warden	Our app will be used by the virtual warden after passing a series of tests. They will record the violations occurring through their mobile camera and upload them to our server through our app. The camera will be fixed in the car, and all they need to do is say a keyword, which will save a piece of the recording and send it to the server where all processing will occur using AI.

Table 1.1: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case Table

Use Case Title	Record and Upload Violation Video with Voice Activation
Actor	Virtual Warden (User)
Description	The user/warden opens the mobile camera, turns on the video recording and fits it on the dashboard of the car. Whenever a user will encounter any violation he will say a word violation and a chunk of video will be recorded focusing the number plate of the car performing violation from 30 seconds earlier till 30 seconds after the violation has been detected. Only that part of the video will be saved on the mobile phone. At the end of the day all the chunks of violated videos will be uploaded on the server for further verification of violation by the higher authorities. If the violation is valid, challan will be generated on the name of the user on the basis of the number plate registration of that user. If the violation is not valid a cut will be generated to the user who uploaded the video because he wasted the time of authorities in checking the violation in that video.
Preconditions	1. User must be logged in. 2. Mobile camera should be fixed properly. 3. Voice should be enabled.
Main Success Scnerio	1. The user opens up the app and starts recording. 2. The system starts recording the front view of the car. 3. The user detects any violation. 4. User says a command/word "violation". 5. The system saves the chunk of violated part from 30 seconds earlier of violation till 30 seconds after the violation have been performed. 6. The process continuous throughout the day whenever a user is using its car. 7. At the end of the day all the chunks of violated videos are uploaded on server.
Postconditions	1. The chunks of videos are successfully uploaded to the server. 2. The AI process the videos and send to authorities to detect traffic violations.
Extentions	1. If the voice command is not working or the app didn't detect your voice command, the user can manually save the chunk of the video via app. 2. If there is no internet connection at the time of upload, the system generates a message of no internet connection and informs the user that video will be uploaded automatically when the connection is restored.
Special Requirements	1. Smooth voice recognition. 2. Enough storage space for storing video chunks in mobile phones. 3. Secure and efficient data upload to the server.

Table 2.1: Use Case

2.2 Functional Requirements

2.2.1 Module 1: App Development & License Plate Detection

This module covers the development of the mobile app, voice command recognition, and the functionality for license plate detection using Optical Character Recognition (OCR).

1. App Development:

(a) Camera Recording:

- i. **FR1.1:** The app shall continuously record the video through the mobile fixed on the dashboard.
- ii. **FR1.2:** The app shall save a chunk of the video being recorded using voice command by the user .
- iii. **FR1.3:** The recorded video shall be saved locally until uploaded to the server.

(b) Voice Command Recognition:

- i. **FR1.4:** The system shall detect specific voice commands (e.g., "Violation") to save the recorded video chunks of traffic violations locally to be uploaded to the server.

(c) UI/UX:

- i. **FR1.6:** The app shall have a user-friendly interface that will allow the user to start and stop recording videos on voice command and also view videos and upload them to the server.
- ii. **FR1.7:** The app shall give voice feedback on saving the video and starting and stopping the video.
- iii. **FR1.8:** The app shall have a dashboard for the user for their records and analytics and record their incentive in detecting violations.

2. License Plate Detection:

(a) OCR Number Plate Detection:

- i. **FR1.9:** The OCR shall be used to get the license plate number of the violator. The license plate shall be captured using the object detection model.
- ii. **FR1.10:**

2.2.2 Module 2: Illegal Number Plate Detection & HTV First Lane Detection

This module focuses on detecting illegal number plates and heavy transport vehicles (HTV) using lane detection and vehicle classification.

(a) Illegal Number Plate Detection:

i. Custom Dataset Collection:

- A. **FR2.1:** The system shall allow the collection of a custom dataset of images containing various types of illegal number plates and number plates of each region
- B. **FR2.2:** The system shall support dataset updates with new images added to the dataset to improve model accuracy time to time.

ii. Model Training:

- A. **FR2.5:** The system shall train a machine learning model using the annotated dataset saved to detect illegal number plates in video footage uploaded to the server.

(b) HTV First Lane Detection:

i. First Lane Detection:

- A. **FR2.7:** The system shall detect the HTV on the first lane by first detecting the first lane and then the HTV on it and then report.

ii. Vehicle Classification:

- A. **FR2.9:** The system shall classify all vehicles using the object detection. And then detect violation based on vehicle type.

iii. Model Training:

- A. **FR2.11:** The system shall train a machine learning model for HTV detection based on a custom dataset.

2.3 Non-Functional Requirements

2.3.1 Reliability

The system shall have a MTBF of at least 200 hours of continuous operation. Our system shall have a downtime of no more than 2 hours for usual maintenance of server and new updates, so we can ensure complete availability of server to user fro working. The recorded videos shall be uploaded to the server without any corruption. Backup shall be kept in the app if any corruption of video occurs.

2.3.2 Usability

The system shall offer an appealing UI that shall allow users to record and upload videos easily. The voice command feature for recording violation video chunks shall have an accuracy rate of at least 90%, and users shall receive clear feedback for successful or failed commands. The app will be very easy to use.

2.3.3 Performance

The mobile app shall function smoothly, ensuring that the video recording and saving of video don't impact the device's performance significantly, maintaining at least 30 FPS for video recording.

2.3.4 Security

All personal data such as videos, user information, and violation reports shall be secured. Unauthorized access to uploaded videos or violation data must be prevented using secure and proper authentication and role-based access controls. The app access will be given to only authorized users.

2.4 Domain Model

Create a representation of the domain model for your project.

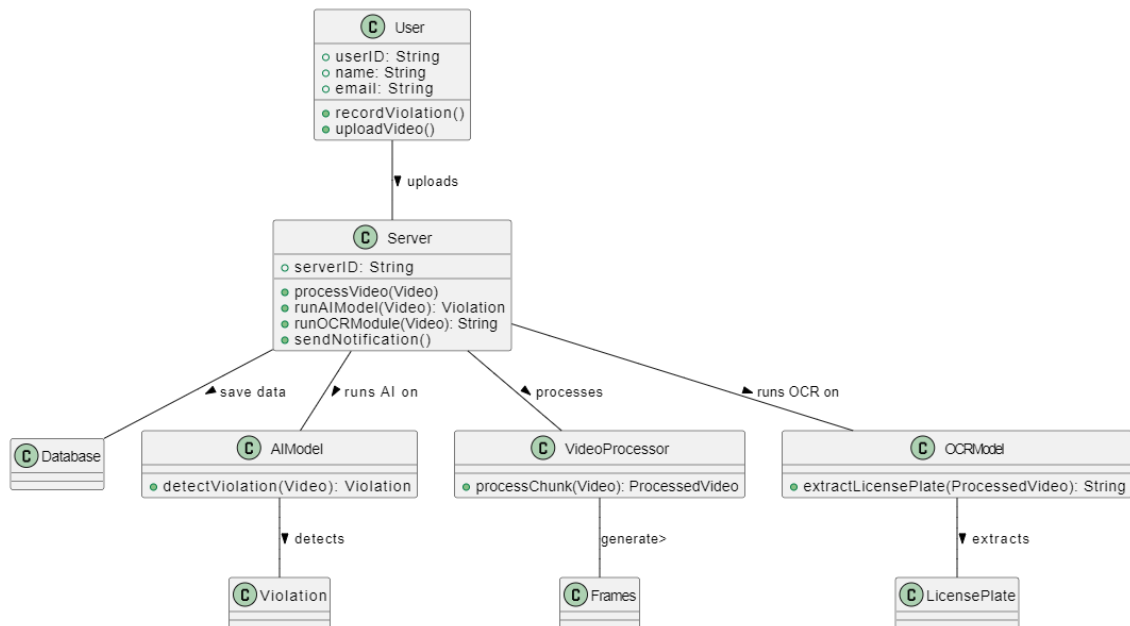


Figure 2.1: Domain Model Diagram

Chapter 3

System Overview

In our App , The user will open our app and fix the device on the dashboard .The device will record the video . When the user comes across a violation he will use voice command to save the chunk of the video in which the violation will be captured . The video chunk will be saved locally in the phone storage. Then the user will upload all captured videos to the server. The server will run an AI model for all violation detection using a pipeline. In the end if any violation detected we will use OCR to get the license plate and notify the authorities and the user .

3.1 Architectural Design

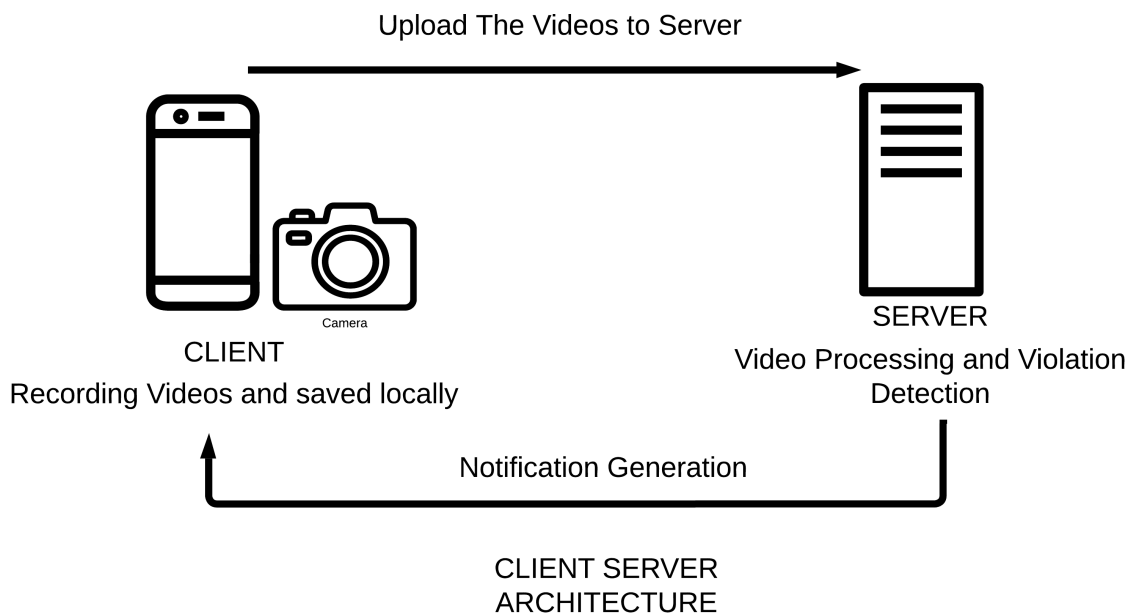


Figure 3.1: Architectural Design Diagram

3.2 Design Models

Design Models for Procedural Approach

3.2.1 Activity Diagram

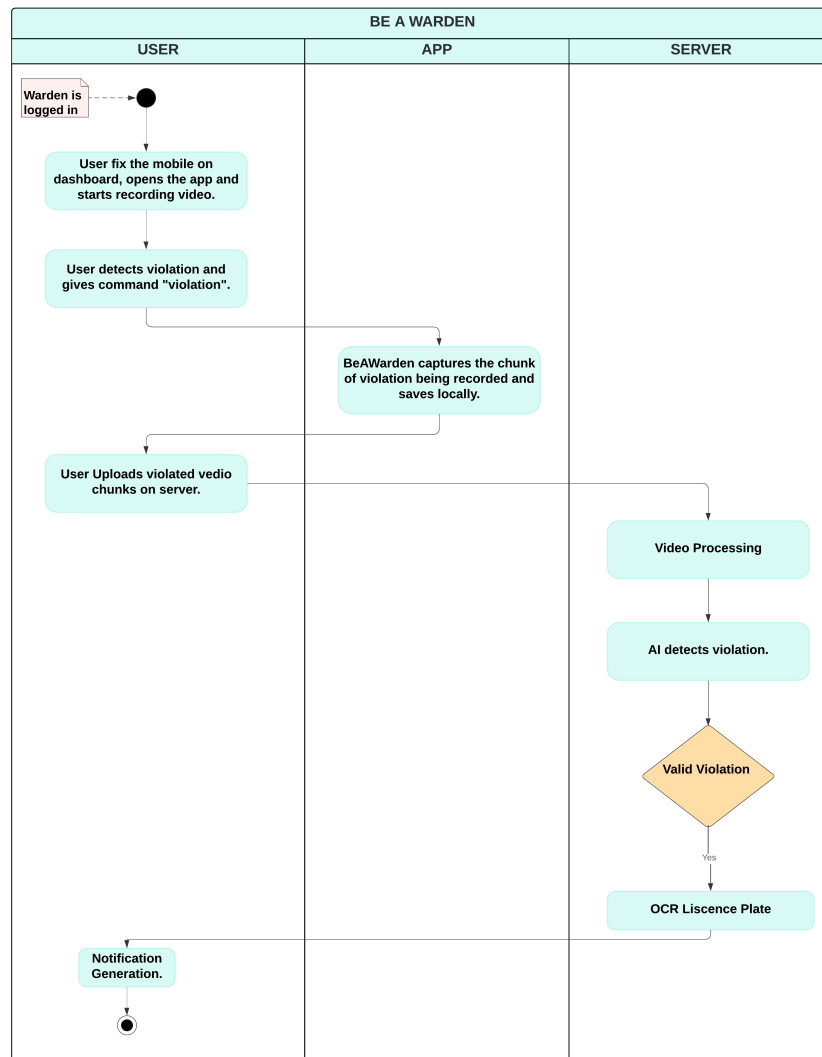


Figure 3.2: Activity Diagram

The activity diagram represents the flow of system. It starts from a precondition that a warden is already logged in. The warden opens up the app starts video recording and fix the mobile phone on the dashboard. Whenever a warden detects violation he says a word "violation" and a chunk of the video detecting violation is saved locally on mobile phone. Rest of the recording is automatically removed from storage as it is of no use. At the end of the day, the warden uploads all the chunks of videos on server where AI confirms that

the violation is valid and generates a challan on the name of user based on registration of number plate of the car.

3.2.2 System Sequence Diagram

The system sequence diagram shows the interactions between external actors and the system during the execution of a use case.

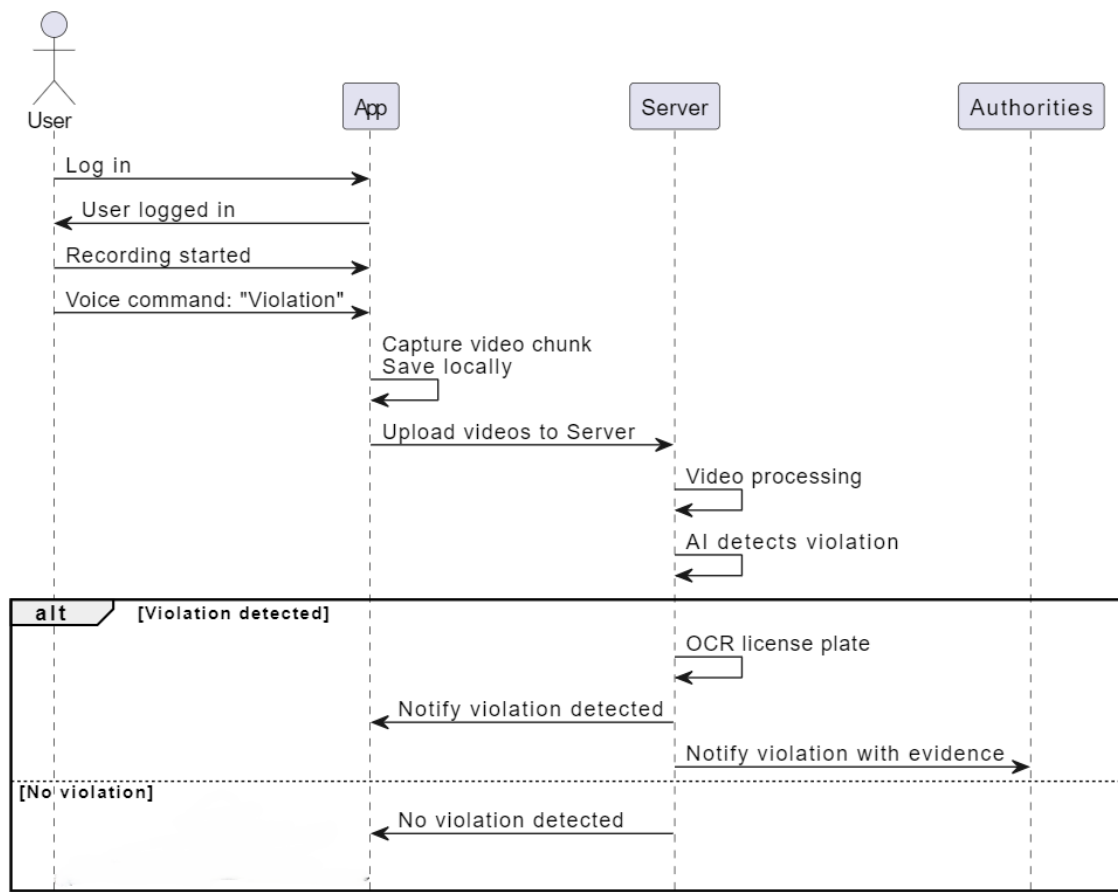


Figure 3.3: System Sequence Diagram

The system comprises of 4 modules

1. Actor
2. Be a Warden App
3. Server
4. Authorities

The actor is a warden that is basically the user of the app. The warden detects the violation and records the video of violation. These videos are uploaded on server by the warden

where validity of violation is checked. Ai checks the validity and confirms that the violation is valid. If it is valid than the chunk and notification is sent to higher authorities so that they can generate challan to the violator.

3.2.3 Data Flow Diagram

The system sequence diagram shows the interactions between external actors and the system during the execution of a use case.

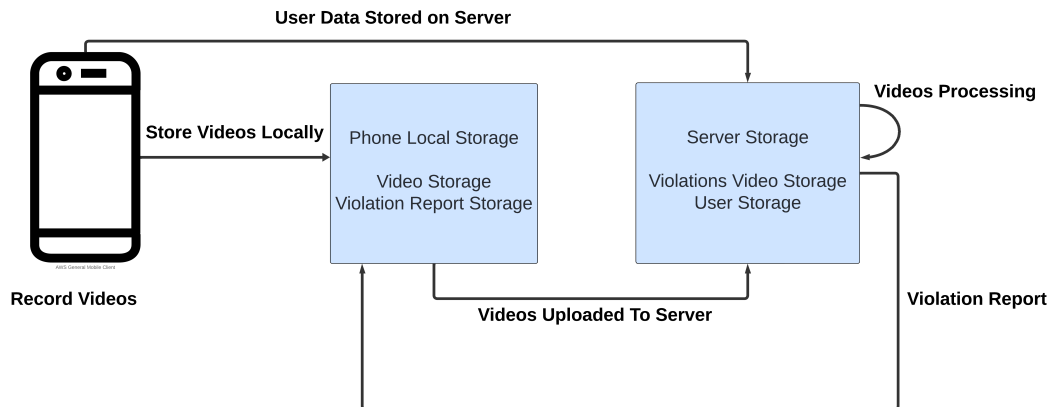


Figure 3.4: Data Flow Diagram

3.3 Data Design

The videos made from the mobile are stored locally on the phone using the phone storage. Then the videos are uploaded to server and hosted there. Then the videos are passed to the AI model stored on the server. The videos on which the AI model is applied are discarded from the server if no violations detected if violation detected then the videos are saved somewhere to send to the user for proof.

3.3.1 Databases

- Firebase
- PostgreSQL
- AWS

Bibliography